

Recitation 1: Review of Statistics

ECON UN3412: Econometrics

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1 Hypothesis Testing

1.1 Two-Sided Test

Consider an unknown population parameter θ (e.g., a population mean μ , a regression coefficient β , or a difference in means). We estimate θ using a sample estimator $\hat{\theta}$ with estimated standard error $\text{SE}(\hat{\theta})$.

Our goal is to test the null hypothesis against the two-sided alternative:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0$$

Standard Error of an Estimator

The **standard error** of an estimator $\hat{\theta}$, denoted $\sigma_{\hat{\theta}}$, is the standard deviation of its sampling distribution:

$$\sigma_{\hat{\theta}} = \sqrt{\text{Var}(\hat{\theta})}$$

Because the true variance $\text{Var}(\hat{\theta})$ typically depends on unknown population parameters, in practice we use the **estimated standard error**, denoted $\text{SE}(\hat{\theta})$ (or $\hat{\sigma}_{\hat{\theta}}$):

$$\text{SE}(\hat{\theta}) = \sqrt{\widehat{\text{Var}}(\hat{\theta})}$$

which substitutes consistent sample estimators for the unknown parameters.

Step-by-Step Testing Procedure

Step 1: Compute the Test Statistic & State Its Null Distribution

The general t -statistic is defined as:

$$t = \frac{\hat{\theta} - \theta_0}{\text{SE}(\hat{\theta})}$$

Under H_0 and standard regularity conditions (i.i.d. sampling, finite variance), by the [Central Limit Theorem \(CLT\)](#) and [Slutsky's Theorem](#), as $n \rightarrow \infty$:

$$t = \frac{\hat{\theta} - \theta_0}{\text{SE}(\hat{\theta})} \xrightarrow{d} \mathcal{N}(0, 1)$$

The statistic converges in distribution to a **standard normal distribution**. In practice, for large samples ($n \geq 30$), critical values and p -values are drawn from $\mathcal{N}(0, 1)$.

Step 2: Determine the Critical Values & Rejection Rule

For a chosen significance level α (e.g., 10%, 5%, 1%), we reject H_0 in favor of H_1 if:

$$|t^{\text{act}}| > c_{\alpha/2}$$

where $c_{\alpha/2}$ is the two-sided critical value such that $\Pr(|Z| > c_{\alpha/2}) = \alpha$.

Under the large-sample normal approximation ($\mathcal{N}(0, 1)$):

Significance Level (α)	Confidence Level		Rejection Rule
	$(1 - \alpha)$	Critical Value ($c_{\alpha/2}$)	
10%	90%	1.645	Reject H_0 if $ t^{\text{act}} > 1.645$
5%	95%	1.960	Reject H_0 if $ t^{\text{act}} > 1.960$
1%	99%	2.576	Reject H_0 if $ t^{\text{act}} > 2.576$

Step 3: Compute and Interpret the p -Value

The p -value is the probability of observing a test statistic at least as extreme as the actual value computed (t^{act}), assuming the null hypothesis H_0 is true:

$$p\text{-value} = 2 \cdot \Pr(Z \geq |t^{\text{act}}|) = 2 \cdot [1 - \Phi(|t^{\text{act}}|)]$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function (CDF).

- **Interpretation:** The p -value measures the plausibility of H_0 given the data. If $p\text{-value} \leq \alpha$, we reject H_0 at significance level α .

Step 4: Construct and Interpret Confidence Intervals

A $(1 - \alpha)$ confidence interval for θ is given by:

$$[\hat{\theta} - c_{\alpha/2} \cdot \text{SE}(\hat{\theta}), \quad \hat{\theta} + c_{\alpha/2} \cdot \text{SE}(\hat{\theta})]$$

- **Interpretation:** In repeated sampling, $(1 - \alpha)\%$ of such constructed intervals will cover the true parameter θ .
- **Duality with Hypothesis Testing:** A two-sided test rejects $H_0 : \theta = \theta_0$ at significance level α if and only if θ_0 lies **outside** the $(1 - \alpha)$ confidence interval.

1.2 One-Sided Test

The test statistic $t = \frac{\hat{\theta} - \theta_0}{\text{SE}(\hat{\theta})}$ and its distribution under H_0 ($\mathcal{N}(0, 1)$) remain **identical** to the two-sided test. What changes is the **alternative hypothesis**, the **rejection region** (all α in one tail), and the **p-value calculation**.

Alternative Hypotheses

- **Right-tailed test:** $H_0 : \theta = \theta_0$ vs. $H_1 : \theta > \theta_0$
- **Left-tailed test:** $H_0 : \theta = \theta_0$ vs. $H_1 : \theta < \theta_0$

Critical Values & Rejection Regions (Entire α in One Tail)

Because all of the rejection probability α is placed in a single tail (instead of being split into $\alpha/2$), we use the one-sided critical value c_α satisfying $\Pr(Z > c_\alpha) = \alpha$:

Significance Level (α)	Critical Value (c_α)	Right-Tailed Rejection Rule ($H_1 : \theta > \theta_0$)	Left-Tailed Rejection Rule ($H_1 : \theta < \theta_0$)
10%	1.282	Reject H_0 if $t^{\text{act}} > 1.282$	Reject H_0 if $t^{\text{act}} < -1.282$
5%	1.645	Reject H_0 if $t^{\text{act}} > 1.645$	Reject H_0 if $t^{\text{act}} < -1.645$
1%	2.326	Reject H_0 if $t^{\text{act}} > 2.326$	Reject H_0 if $t^{\text{act}} < -2.326$

p-Value Calculation (No Factor of 2)

We do **not** multiply by 2 because we only accumulate probability mass in the direction of the alternative:

- **Right-tailed ($H_1 : \theta > \theta_0$):**

$$p\text{-value} = \Pr(Z \geq t^{\text{act}}) = 1 - \Phi(t^{\text{act}})$$

- **Left-tailed ($H_1 : \theta < \theta_0$):**

$$p\text{-value} = \Pr(Z \leq t^{\text{act}}) = \Phi(t^{\text{act}})$$

One-Sided Confidence Bounds

Rather than a two-sided interval, a one-sided test corresponds to a **one-sided confidence bound** at level $(1 - \alpha)$:

- **Lower bound** ($H_1 : \theta > \theta_0$):

$$[\hat{\theta} - c_\alpha \cdot \text{SE}(\hat{\theta}), \quad \infty)$$

- **Upper bound** ($H_1 : \theta < \theta_0$):

$$(-\infty, \quad \hat{\theta} + c_\alpha \cdot \text{SE}(\hat{\theta})]$$

1.3 Equivalence of the Three Testing Methods

In practice, hypothesis testing is often presented through three different “methods”:

1. **Critical Value Approach:** Reject H_0 if $|t^{\text{act}}| > c_{\alpha/2}$ (or $t^{\text{act}} > c_\alpha$ for one-sided).
2. **p -Value Approach:** Reject H_0 if $p\text{-value} \leq \alpha$.
3. **Confidence Interval Approach:** Reject H_0 if $\theta_0 \notin \text{CI}_{1-\alpha}$.

Key Takeaway: These three methods are **mathematically identical**. At any chosen significance level α , they will **always** yield the exact same conclusion (reject H_0 vs. fail to reject H_0).

Why They Are Mathematically Identical

Starting from the confidence interval condition:

$$\begin{aligned} \theta_0 \notin [\hat{\theta} - c_{\alpha/2} \cdot \text{SE}(\hat{\theta}), \hat{\theta} + c_{\alpha/2} \cdot \text{SE}(\hat{\theta})] &\iff |\hat{\theta} - \theta_0| > c_{\alpha/2} \cdot \text{SE}(\hat{\theta}) \\ &\iff \frac{|\hat{\theta} - \theta_0|}{\text{SE}(\hat{\theta})} > c_{\alpha/2} \\ &\iff |t^{\text{act}}| > c_{\alpha/2} \quad (\text{Critical Value Rule}) \\ &\iff 2[1 - \Phi(|t^{\text{act}}|)] \leq 2[1 - \Phi(c_{\alpha/2})] \\ &\iff p\text{-value} \leq \alpha \quad (p\text{-Value Rule}) \end{aligned}$$

How Their Interpretations Differ

While the mathematical decision is identical, each method provides a distinct perspective:

Method	Question It Answers	Key Advantage
Critical Value	<i>“Is the test statistic further into the tail than the critical threshold $c_{\alpha/2}$?”</i>	Simple binary decision rule based directly on standard deviations from the null.

Method	Question It Answers	Key Advantage
<i>p</i>-Value	<i>“Assuming H_0 is true, what is the probability of observing an estimate at least as extreme as $\hat{\theta}$?”</i>	Continuous measure of evidence strength; allows readers to evaluate the test at any significance level α .
Confidence Interval	<i>“Is the hypothesized value θ_0 among the set of plausible values for the true parameter θ?”</i>	Retains original economic units; simultaneously displays estimation uncertainty, statistical significance, and economic magnitude.

2 Multiple Random Variables

Consider two random variables X and Y . In econometrics, we analyze pairs of variables simultaneously (e.g., education and wages, class size and test scores).

- **Discrete Random Variables:** Take on distinct, countable values. Characterized by **probability mass functions (PMF)** and summations ($\sum$).
- **Continuous Random Variables:** Take on values across a continuous range. Characterized by **probability density functions (PDF)** and integrals ($\int$).

2.1 Joint Probability Distribution

The **joint distribution** models the simultaneous behavior of X and Y :

- **Discrete (Joint PMF):**

$$\Pr(X = x, Y = y)$$

Satisfies $\Pr(X = x, Y = y) \geq 0$ and total probability:

$$\sum_x \sum_y \Pr(X = x, Y = y) = 1$$

- **Continuous (Joint PDF):** A joint probability density function $f_{X,Y}(x, y) \geq 0$ such that:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

The probability that (X, Y) falls in a region A is:

$$\Pr((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

2.2 Marginal Probability Distribution

The **marginal distribution** of a single variable is obtained by integrating or summing out the other variable:

Marginal PMF / PDF

- **Discrete:**

$$\Pr(X = x) = \sum_y \Pr(X = x, Y = y) \quad \text{and} \quad \Pr(Y = y) = \sum_x \Pr(X = x, Y = y)$$

- **Continuous:**

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

Marginal Expectation (Mean) and Variance

- **Discrete:**

$$\mu_Y = E[Y] = \sum_y y \cdot \Pr(Y = y)$$

$$\sigma_Y^2 = \text{Var}(Y) = \sum_y (y - \mu_Y)^2 \Pr(Y = y)$$

- **Continuous:**

$$\mu_Y = E[Y] = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy$$

$$\sigma_Y^2 = \text{Var}(Y) = \int_{-\infty}^{\infty} (y - \mu_Y)^2 f_Y(y) dy$$

2.3 Conditional Probability Distribution

The **conditional distribution** of Y given that $X = x$ isolates the behavior of Y within a specific subpopulation defined by $X = x$:

- **Discrete:** For $\Pr(X = x) > 0$:

$$\Pr(Y = y \mid X = x) = \frac{\Pr(X = x, Y = y)}{\Pr(X = x)}$$

- **Continuous:** For $f_X(x) > 0$:

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

Conditional Expectation (Conditional Mean)

The expected value of Y given $X = x$ is:

- **Discrete:**

$$E[Y | X = x] = \sum_y y \cdot \Pr(Y = y | X = x)$$

- **Continuous:**

$$E[Y | X = x] = \int_{-\infty}^{\infty} y \cdot f_{Y|X}(y | x) dy$$

Important Concept: For a specific number x , $E[Y | X = x]$ is a deterministic constant. However, considered across the random variable X , the conditional expectation $E[Y | X]$ is itself a **random variable** (a function of X).

2.4 Law of Iterated Expectations (LIE)

The [Law of Iterated Expectations](#) states that the expectation of the conditional expectation of Y given X equals the unconditional expectation of Y :

$$E[Y] = E_X[E_{Y|X}[Y | X]]$$

Proof

Discrete Case:

$$\begin{aligned} E_X[E[Y | X]] &= \sum_x E[Y | X = x] \cdot \Pr(X = x) \\ &= \sum_x \left(\sum_y y \cdot \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} \right) \Pr(X = x) \\ &= \sum_y y \sum_x \Pr(X = x, Y = y) = \sum_y y \cdot \Pr(Y = y) = E[Y] \end{aligned}$$

Continuous Case:

$$\begin{aligned}
E_X[E[Y | X]] &= \int_{-\infty}^{\infty} E[Y | X = x] \cdot f_X(x) dx \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} y \cdot \frac{f_{X,Y}(x, y)}{f_X(x)} dy \right) f_X(x) dx \\
&= \int_{-\infty}^{\infty} y \left(\int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \right) dy = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy = E[Y]
\end{aligned}$$

2.5 Statistical Independence

Two random variables X and Y are **statistically independent** if the distribution of one does not depend on the value of the other:

- **Discrete:** $\Pr(X = x, Y = y) = \Pr(X = x) \cdot \Pr(Y = y)$ for all x, y
- **Continuous:** $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$ for all x, y

Consequences of Independence:

$$E[Y | X] = E[Y], \quad E[XY] = E[X]E[Y], \quad \text{and} \quad \text{Cov}(X, Y) = 0$$

2.6 Summary: Discrete vs. Continuous Counterparts

Concept	Discrete Random Variables	Continuous Random Variables
Joint Distribution	$\Pr(X = x, Y = y)$	$f_{X,Y}(x, y)$
Total Probability	$\sum_x \sum_y \Pr(X = x, Y = y) = 1$	$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$
Marginal Distribution	$\Pr(X = x) = \sum_y \Pr(X = x, Y = y)$	$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$
Conditional Distribution	$\Pr(Y = y X = x) = \frac{\Pr(X=x, Y=y)}{\Pr(X=x)}$	$f_{Y X}(y x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$
Conditional Mean	$E[Y X = x] = \sum_y y \Pr(Y = y X = x)$	$E[Y X = x] = \int_{-\infty}^{\infty} y f_{Y X}(y x) dy$
Expectation Operator	$E[g(X)] = \sum_x g(x) \Pr(X = x)$	$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$

3 Measures of Association

3.1 Covariance

The **covariance** measures the direction of the linear relationship between two random variables X and Y .

Population Covariance

$$\text{Cov}(X, Y) = \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X\mu_Y$$

- $\text{Cov}(X, Y) > 0$: X and Y tend to move in the same direction.
- $\text{Cov}(X, Y) < 0$: X and Y tend to move in opposite directions.
- $\text{Cov}(X, Y) = 0$: X and Y have no linear association (they are uncorrelated).

The Units Problem of Covariance

The unit of $\text{Cov}(X, Y)$ is the **product of the units of X and Y** (e.g., if X is measured in dollars and Y in hours, $\text{Cov}(X, Y)$ is in dollar-hours). Consequently, the numerical magnitude of covariance depends on the scale of measurement and cannot be used to judge the *strength* of an association.

Sample Covariance

Given sample observations $(X_1, Y_1), \dots, (X_n, Y_n)$:

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

3.2 Correlation Coefficient

The **correlation coefficient** standardizes the covariance by dividing by the product of the standard deviations, creating a unit-free measure of linear association.

Population Correlation

$$\text{Corr}(X, Y) = \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

Key Properties

1. **Bounded Range:** By the **Cauchy-Schwarz inequality**:

$$-1 \leq \rho_{XY} \leq 1$$

2. **Interpretation of Extremes:**

- $\rho_{XY} = 1$: Perfect positive linear association ($Y = a + bX$ with $b > 0$).
- $\rho_{XY} = -1$: Perfect negative linear association ($Y = a + bX$ with $b < 0$).
- $\rho_{XY} = 0$: No linear relationship (X and Y are uncorrelated).

3. **Scale Invariance:** ρ_{XY} is unaffected by multiplying or adding constants:

$$\text{Corr}(a + bX, c + dY) = \text{sign}(bd) \cdot \text{Corr}(X, Y)$$

4. Independence vs. Uncorrelatedness:

$$\text{Independence} \implies \text{Corr}(X, Y) = 0$$

*(The converse is generally not true: zero correlation only implies no **linear** relationship; non-linear dependencies may still exist.)*

Sample Correlation

$$r_{XY} = \frac{S_{XY}}{S_X S_Y} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$$
